DriveCompose: Compositional Generative Reconstruction of Driving Scenes via Joint Completion

Supplementary Material

A. Cross-Dataset Generalization

We evaluate zero-shot transfer from Waymo to nuScenes to assess reconstruction quality beyond the training dataset. The Waymo-trained checkpoint is applied directly to nuScenes, with all model parameters fixed during evaluation.

As shown in Table S1, DriveCompose achieves the highest PSNR and SSIM among the compared methods. It reaches 26.47 dB PSNR and 0.803 SSIM, improving over the strongest baseline, DGGT, by 1.16 dB and 0.009, respectively. These results demonstrate improved reconstruction quality on the target dataset without target-domain adaptation.

Table S1. Zero-shot generalization from Waymo to nuScenes. Higher PSNR and SSIM are better.

	Zero-shot	
Method	PSNR ↑	SSIM ↑
STORM [4]	17.77	0.669
DGGT [1]	25.31	0.794
StreetForward [5]	23.17	0.712
Ours	**26.47**	**0.803**

B. Training details

Learning is divided into four stages so that the completion target has a stable meaning. Recovery determines the evidence and support, the codec defines its latent coordinates, and the completion network learns a distribution over the missing codes. The last stage adapts this distribution to real observations without changing the representation it predicts.

Stage 1: visual recovery. We supervise the image encoder, geometry heads, identity slots, and state heads using geometry, semantics, instance assignments, and actor annotations. One trajectory-level matching assigns slots to target identities and is reused at every timestamp. Root translation, rotation, and fixed dimensions constrain rigid-asset states; humans additionally require identity-consistent shape, template binding, and joint supervision. Confidence is trained or calibrated against geometric reliability, using depth, occlusion, and reprojection information. These targets establish the coordinate and identity consistency required by aggregation.

Stage 2: latent representation fitting. We freeze recovery and train the type-specific encoders and decoders. For each sample, partial input establishes the query layout once. More complete observations are then aggregated on exactly the same positions, types, order, and ownership. This shared layout aligns partial evidence and complete targets query by query. RGB, depth, and silhouette supervision train the decoded scene, with empty-space opacity and Gaussian-scale regularization. Gaussian-set targets can also be matched as sets; direct coordinate supervision is used only when child correspondences are available.

For rendered image $\widehat{\mathbf{I}}$ and depth $\widehat{d}$, the observation terms used in representation fitting and later adaptation are masked L1 errors:

$$\mathcal{L}_{\text{rgb}} = \langle \|\widehat{\mathbf{I}} - \mathbf{I}\|_1/3 \rangle_{\Omega_I}, \tag{S1}$$

$$\mathcal{L}_{\text{depth}} = \langle |\widehat{d} - d| \rangle_{\Omega_d}. \tag{S2}$$

The notation $\langle \ell \rangle_\Omega$ denotes a mean over valid elements of mask Ω. The depth target d and its validity mask are supplied by the available supervision.

Stage 3: completion-prior learning. We freeze recovery and the entire codec, encode complete targets as $\mathbf{z}_1$, and optimize the completion transformer with Eq. (9) of the main paper. Complete scene or asset examples provide target codes for the unknown queries in Eq. (8) of the main paper. Complete scenes provide cross-component context, while complete assets supply within-object geometry and appearance.

Conditioning masks simulate contiguous unseen surfaces, background holes, long occlusions, short tracks, and uneven camera coverage. Perturbations to recovered poses, instance assignments, and confidence expose the generator to imperfect evidence. Target codes and input codes retain the same codec and query definition throughout this stage.

Stage 4: adaptation through rendering. Keeping recovery and the codec frozen, we differentiate through flow integration, decoding, and rendering to update only the completion network. Partial real logs supervise the rendered observations. Known-free rays and geometric backend samples constrain added content:

$$\mathcal{L}_{\text{scene}} = \lambda_e \langle a_r^2 \rangle + \lambda_g \langle [-d_f]_+^2 \rangle + \lambda_c \langle |d_f| \rangle_{\Omega_c} + \lambda_p \langle [d_p]_+^2 \rangle, \tag{S3}$$

$$\mathcal{L}_{\text{adapt}} = \lambda_r \mathcal{L}_{\text{rgb}} + \lambda_d \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{scene}}. \tag{S4}$$

Here a_r is accumulated opacity along a known-free ray, d_f is foot signed distance positive above ground, Ω_c selects contacting feet, and d_p is penetration depth positive inside another actor. We use $[x]_+ = \max(x, 0)$; unmasked means run over supplied samples. The coefficients λ weight the active constraints. Free-ray opacity constrains the rendered effects of added content beyond the query-level free-space mask.

Complete-target samples can be interleaved with real-log adaptation to retain the learned prior, yielding $\mathcal{L}_4 = \lambda_f \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{adapt}}$, with each term applied only to samples carrying its required targets. Observed anchors are enforced by frozen decoding and hard code restoration.

Inference. A log is encoded once, its support and evidence are fixed, and the completion model integrates the unknown codes. Type-specific decoders produce the background and canonical assets. Rendering composes these fixed components under the requested camera and actor states.

C. Scene composition details

State-driven rendering. At online step k, state s_k specifies the ego pose, active assets, rigid transforms, and human root and joint poses. A rigid asset's Gaussian center and covariance transform as

$$\boldsymbol{\mu}^w = \mathbf{R}\boldsymbol{\mu}^a + \mathbf{t}, \qquad \boldsymbol{\Sigma}^w = \mathbf{R}\boldsymbol{\Sigma}^a\mathbf{R}^\top. \tag{S5}$$

For humans, the weighted skinning transform is $\mathbf{A}_i = \sum_j w_{ij}\mathbf{P}_j\mathbf{B}_j^{-1}$, where $\mathbf{B}_j$ and $\mathbf{P}_j$ are rest and current global joints in the root frame. The transformed center is $\bar{\boldsymbol{\mu}}_i^w = \mathbf{T}_h^w\mathbf{A}_i\bar{\boldsymbol{\mu}}_i^h$. The combined local linear map $\mathbf{J}_i$ gives covariance $\mathbf{J}_i\boldsymbol{\Sigma}_i\mathbf{J}_i^\top$. Its inverse $\mathbf{J}_i^{-1}$ maps viewing directions into the canonical frame; the direction is normalized before appearance evaluation. The inverse is evaluated for nonsingular skinning transforms. The root pose controls overall placement, while the root joint retains its rest transform.

The renderer composites the background, active assets, and valid sky. External policy and dynamics can advance the state:

$$o_k = \mathcal{R}(\text{Compose}(\mathcal{W}, s_k), \mathcal{C}_k), \tag{S6}$$

$$s_{k+1} = F_{\text{dyn}}(s_k, \pi(o_k), \Delta t). \tag{S7}$$

Camera and actor state changes reuse the same background and canonical assets. Road meshes and collision proxies use the same coordinates as the visual components, allowing their motion to be driven by a physical simulator.

D. Visual recovery and aggregation details

Input conventions and preprocessing. The input is a synchronized log of images $\mathbf{I}_{t,v}$, timestamps, intrinsics $\mathbf{K}_v$, and camera-to-rig calibration $\mathbf{T}_{c_v}^r$. Rig poses $\mathbf{T}_r^w(t)$ are provided or estimated relative to the first frame, giving camera poses $\mathbf{T}_{c_v}^w(t) = \mathbf{T}_r^w(t)\mathbf{T}_{c_v}^r$. Transforms act on column vectors and distances are in meters. Images follow a rectified pinhole model and linear RGB; distortion, rolling shutter, and exposure differences are handled before evidence aggregation.

Shared visual encoding. The Driving Log Encoder uses patch tokenization [2] and transformer attention [3], with image-position, camera-calibration, and time embeddings. Within-image attention extracts spatial features. A second transformer processes pooled image summaries across views and timestamps, and returns the resulting context to the patch features. Geometry heads predict camera-axis depth, normals, confidence, semantics, and instance assignments. The confidence determines which predictions can contribute surface evidence.

Learned actor queries attend to all image tokens to form trajectory-level identity slots. Each slot then reads the features at an individual timestamp to estimate its state. This separates persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Root poses are predicted in the rig frame and mapped to world coordinates with $\mathbf{T}_r^w(t)$. A human shape code selects an identity-specific template with fixed canonical support, skeleton topology, and binding. Slot capacity bounds the number of represented actors.

Query supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}_i^w$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; rigid anchors are aligned by inverse SE(3) transforms and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout remains fixed during completion. We obtain $\mathbf{q}_i^w$ by placing the query in the first input configuration, including skinning for humans. Canonical coordinates identify the content within its component; reference-world coordinates let the generator relate that component to its surroundings.

Motion-aware evidence aggregation. For each observation, we place the canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}_a^w(t)\bar{\mathbf{q}}_i), & k_i = \text{rigid}, \\ \text{xyz}(\mathbf{T}_h^w(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{S8}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection into a camera gives depth $z_{i,t,v}$ and sampled feature, depth, and confidence $(\boldsymbol{\phi}_{i,t,v}, d_{i,t,v}, c_{i,t,v})$. With tolerance $\eta = \eta_{\text{abs}} + \eta_{\text{rel}} d$, define $r = (d - z)/\eta$. A surface observation must lie inside the image, have valid depth and sufficient confidence, match the query's instance, and satisfy $|r| \leq 1$. Background queries additionally require static-background semantics. Let $a_{i,t,v}$ denote this validity test. We aggregate

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{S9}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{S10}$$

The query confidence is the maximum over accepted observations. The test excludes observations whose depth or instance identity disagrees with the query. A reliable ray votes for free space only when the entire support cell is in front of the measured surface:

$$z_{i,t,v} + \tfrac{1}{2}\|\boldsymbol{\delta}_i\|_2 < d_{i,t,v} - \eta. \tag{S11}$$

Positive accumulated observation weight marks a query as observed. Otherwise, sufficient free-space votes mark it free, and the remainder is unknown. The masks m_i, f_i, and u_i are mutually exclusive and satisfy $m_i + f_i + u_i = 1$. Sky projection uses camera rotation alone, reliable sky semantics, and zero depth residual; sky queries receive no free-space votes.

E. Gaussian decoding details

A Gaussian stores its center $\boldsymbol{\mu}$, covariance $\boldsymbol{\Sigma}$, opacity α, and directional appearance.

Background, rigid, and human Gaussian decoders independently map each query to K child Gaussians. For child k of query i, the center is confined to its support cell:

$$\boldsymbol{\mu}_{i,k} = \mathbf{q}_i + \tfrac{1}{2}\boldsymbol{\delta}_i \odot \tanh(\mathbf{b}_{i,k}). \tag{S12}$$

Positive scales and normalized rotations determine positive-definite covariance. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. Human children inherit the skinning weights of their support query; rigid children share their asset's rigid transform. A separate sky head predicts cubemap colors, with the observation-supported domain retained in V_{sky}.

Appearance uses a constant RGB term and three local directional coefficients per channel, evaluated as $c(d) = \max(0, c_0 + C_{\text{app}} \hat{d}_{\text{local}})$. Decoding is per query: once the codec is frozen, an unchanged observed code produces unchanged Gaussian parameters. The completion network can exchange context across queries without changing this decoding property.

References

[1] Xiaoxue Chen, Ziyi Xiong, Yuantao Chen, Gen Li, Nan Wang, Hongcheng Luo, Long Chen, Haiyang Sun, Bing Wang, Guang Chen, Hongyang Li, Ya-Qin Zhang, Hangjun Ye, and Hao Zhao. DGGT: Feedforward 4D Reconstruction of Dynamic Driving Scenes using Unposed Images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1265–1276, 2026. 1

[2] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*, 2021. 2

[3] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is All you Need. In *Advances in Neural Information Processing Systems*, 2017. 2

[4] Jiawei Yang, Jiahui Huang, Boris Ivanovic, Yuxiao Chen, Yan Wang, Boyi Li, Yurong You, Apoorva Sharma, Maximilian Igl, Peter Karkus, Danfei Xu, Yue Wang, and Marco Pavone. STORM: Spatio-TempOral Reconstruction Model For Large-Scale Outdoor Scenes. In *International Conference on Learning Representations*, pages 50446–50465, 2025. 1

[5] Zhongrui Yu, Zhao Wang, Yijia Xie, Yida Wang, Xueyang Zhang, Yifei Zhan, and Kun Zhan. StreetForward: Perceiving Dynamic Street with Feedforward Causal Attention. *arXiv preprint arXiv:2603.19552*, 2026. 1